



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 07/11/2018*

EXERCISE

We observe N samples of a realization of the random process $X[n] = A + B \sin(2\pi F_0 n) + W[n]$, for $n=0, 1, \dots, N-1$, where $N \gg 1$. A and B are unknown deterministic parameters and F_0 is the known digital frequency. $\{W[n]; n = 0, 1, \dots, N-1\}$ are independent and identically distributed (IID) random variables modeling additive noise; each noise sample is Gaussian distributed, with zero mean and known variance σ_w^2 .

- (1) Derive the joint maximum likelihood (ML) estimator of the unknown parameters A and B .
- (2) Derive the bias and the mean square error (MSE) of the ML estimator of A and determine whether the estimator is unbiased and consistent. Then, plot the MSE of $\hat{A}_{ML}$ as a function of N and derive the value of N which guarantees an MSE lower than 10^{-3} when the noise power is $\sigma_w^2 = 1$.
- (3) Derive the bias and the mean square error (MSE) of the ML estimator of B .
- (4) Derive the CRBs on the estimation of A and B and determine if the two estimators are efficient.



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Solution:

(1) Derive the joint maximum likelihood (ML) estimator of the unknown parameters A and B .

$$X[n] = A + B \sin(2\pi F_0 n) + W[n], \quad n = 0, 1, \dots, N-1$$

$$\mathbf{X} = A\mathbf{1} + B\mathbf{s} + \mathbf{W} \in N(A\mathbf{1} + B\mathbf{s}, \sigma_w^2 \mathbf{I})$$

where:

$$\mathbf{1} \triangleq \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \quad \mathbf{s} \triangleq \begin{bmatrix} 0 \\ \sin(2\pi F_0) \\ \vdots \\ \sin(2\pi F_0(N-1)) \end{bmatrix}, \quad \Rightarrow \quad \|\mathbf{1}\|^2 = N, \quad \|\mathbf{s}\|^2 = N/2, \quad \mathbf{1}^T \mathbf{s} = 0 \quad (\text{if } N \gg 1)$$

The data vector $\mathbf{X}$ can be expressed as:

$$\mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{s} \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}, \quad \text{where } \mathbf{H} \triangleq_{N \times 2} \begin{bmatrix} \mathbf{1} & \mathbf{s} \end{bmatrix} \text{ and } \boldsymbol{\theta} \triangleq_{2 \times 1} \begin{bmatrix} A \\ B \end{bmatrix}$$

Since the noise is AWGN, the ML estimator coincide with the LS estimator:

$$\hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} \hat{A}_{ML} \\ \hat{B}_{ML} \end{bmatrix} = \hat{\boldsymbol{\theta}}_{LS} = \mathbf{H}^\# \mathbf{X} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{X}$$

$$\mathbf{H}^T \mathbf{H} = \begin{bmatrix} \mathbf{1}^T \\ \mathbf{s}^T \end{bmatrix} \begin{bmatrix} \mathbf{1} & \mathbf{s} \end{bmatrix} = \begin{bmatrix} \mathbf{1}^T \mathbf{1} & \mathbf{1}^T \mathbf{s} \\ \mathbf{s}^T \mathbf{1} & \mathbf{s}^T \mathbf{s} \end{bmatrix} = \begin{bmatrix} N & 0 \\ 0 & N/2 \end{bmatrix} \Rightarrow (\mathbf{H}^T \mathbf{H})^{-1} = \begin{bmatrix} N & 0 \\ 0 & N/2 \end{bmatrix}^{-1} = \begin{bmatrix} 1/N & 0 \\ 0 & 2/N \end{bmatrix} = \frac{1}{N} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} \hat{A}_{ML} \\ \hat{B}_{ML} \end{bmatrix} = \hat{\boldsymbol{\theta}}_{LS} = \frac{1}{N} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \mathbf{1}^T \\ \mathbf{s}^T \end{bmatrix} \mathbf{X} = \frac{1}{N} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \mathbf{1}^T \mathbf{X} \\ \mathbf{s}^T \mathbf{X} \end{bmatrix} = \frac{1}{N} \begin{bmatrix} \mathbf{1}^T \mathbf{X} \\ 2\mathbf{s}^T \mathbf{X} \end{bmatrix}$$

$$\hat{A}_{ML} = \frac{1}{N} \mathbf{1}^T \mathbf{X} = \frac{1}{N} \sum_{n=0}^{N-1} X[n]$$

$$\hat{B}_{ML} = \frac{2}{N} \mathbf{s}^T \mathbf{X} = \frac{2}{N} \sum_{n=0}^{N-1} X[n] \sin(2\pi F_0 n)$$



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(2) Derive the bias and the mean square error (MSE) of the ML estimator of A and determine whether the estimator is unbiased and consistent. Then, plot the MSE of $\hat{A}_{ML}$ as a function of N and derive the value of N which guarantees an MSE lower than 10^{-3} when the noise power is $\sigma_w^2 = 1$.

The MSE of the two estimators can be derived in different ways. One is the following.

$$\hat{A}_{ML} = \frac{1}{N} \mathbf{1}^T \mathbf{X} = \frac{1}{N} \mathbf{1}^T (A\mathbf{1} + B\mathbf{s} + \mathbf{W}) = \frac{A}{N} \mathbf{1}^T \mathbf{1} + \frac{B}{N} \mathbf{1}^T \mathbf{s} + \frac{1}{N} \mathbf{1}^T \mathbf{W} = A + \frac{1}{N} \mathbf{1}^T \mathbf{W}$$

$$\varepsilon_A = A - \hat{A}_{ML} = -\frac{1}{N} \mathbf{1}^T \mathbf{W} \Rightarrow b\{\hat{A}_{ML}\} = E\{\varepsilon_A\} = E\left\{\left(-\frac{1}{N} \mathbf{1}^T \mathbf{W}\right)\right\} = -\frac{1}{N} \mathbf{1}^T E\{\mathbf{W}\} = 0 \quad (\text{unbiased})$$

$$MSE\{\hat{A}_{ML}\} = E\{\varepsilon_A^2\} = E\left\{\left(-\frac{1}{N} \mathbf{1}^T \mathbf{W}\right)\left(-\frac{1}{N} \mathbf{1}^T \mathbf{W}\right)^T\right\} = \frac{1}{N^2} \mathbf{1}^T E\{\mathbf{W}\mathbf{W}^T\} \mathbf{1} = \frac{\sigma_w^2}{N} \quad (\text{consistent})$$

$$\sigma_w^2 = 1 \Rightarrow MSE\{\hat{A}_{ML}\} = \frac{1}{N} < 10^{-3} \Rightarrow N > 10^3$$

(3) Derive the bias and the mean square error (MSE) of the ML estimator of B .

$$\hat{B}_{ML} = \frac{2}{N} \mathbf{s}^T \mathbf{X} = \frac{2}{N} \mathbf{s}^T (A\mathbf{1} + B\mathbf{s} + \mathbf{W}) = \frac{2A}{N} \mathbf{s}^T \mathbf{1} + \frac{2B}{N} \mathbf{s}^T \mathbf{s} + \frac{2}{N} \mathbf{s}^T \mathbf{W} = B + \frac{2}{N} \mathbf{s}^T \mathbf{W}$$

$$\varepsilon_B = B - \hat{B}_{ML} = -\frac{2}{N} \mathbf{s}^T \mathbf{W} \Rightarrow b\{\hat{B}_{ML}\} = E\{\varepsilon_B\} = E\left\{\left(-\frac{2}{N} \mathbf{s}^T \mathbf{W}\right)\right\} = -\frac{2}{N} \mathbf{s}^T E\{\mathbf{W}\} = 0 \quad (\text{unbiased})$$

$$MSE\{\hat{B}_{ML}\} = E\{\varepsilon_B^2\} = E\left\{\left(-\frac{2}{N} \mathbf{s}^T \mathbf{W}\right)\left(-\frac{2}{N} \mathbf{s}^T \mathbf{W}\right)^T\right\} = \frac{4}{N^2} \mathbf{s}^T E\{\mathbf{W}\mathbf{W}^T\} \mathbf{s} = \frac{2\sigma_w^2}{N} \quad (\text{consistent})$$



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(4) Derive the CRBs on the estimation of A and B and determine if the two estimators are efficient.

The elements of the FIM for the estimation of the parameters of a deterministic signal in AWGN are given by:

$$[\mathbf{I}(\boldsymbol{\theta})]_{i,j} = \frac{1}{\sigma_w^2} \sum_{k=0}^{N-1} \frac{\partial s_k(\boldsymbol{\theta})}{\partial \theta_i} \cdot \frac{\partial s_k(\boldsymbol{\theta})}{\partial \theta_j}$$

In our case, $s_k(\boldsymbol{\theta}) = A + B \sin(2\pi F_0 k)$, so we immediately get:

$$\frac{\partial s_k(\boldsymbol{\theta})}{\partial A} = 1, \quad \frac{\partial s_k(\boldsymbol{\theta})}{\partial B} = \sin(2\pi F_0 k)$$

$$[\mathbf{I}(\boldsymbol{\theta})]_{1,1} = \frac{1}{\sigma_w^2} \sum_{k=0}^{N-1} \frac{\partial s_k(\boldsymbol{\theta})}{\partial A} \cdot \frac{\partial s_k(\boldsymbol{\theta})}{\partial A} = \frac{1}{\sigma_w^2} \sum_{k=0}^{N-1} 1 \cdot 1 = \frac{N}{\sigma_w^2}$$

$$[\mathbf{I}(\boldsymbol{\theta})]_{1,2} = [\mathbf{I}(\boldsymbol{\theta})]_{2,1} = \frac{1}{\sigma_w^2} \sum_{k=0}^{N-1} \frac{\partial s_k(\boldsymbol{\theta})}{\partial A} \cdot \frac{\partial s_k(\boldsymbol{\theta})}{\partial B} = \frac{1}{\sigma_w^2} \sum_{k=0}^{N-1} \sin(2\pi F_0 k) = \frac{N}{\sigma_w^2} \left(\frac{1}{N} \sum_{k=0}^{N-1} \sin(2\pi F_0 k) \right) = 0$$

$$[\mathbf{I}(\boldsymbol{\theta})]_{2,2} = \frac{1}{\sigma_w^2} \sum_{k=0}^{N-1} \frac{\partial s_k(\boldsymbol{\theta})}{\partial B} \cdot \frac{\partial s_k(\boldsymbol{\theta})}{\partial B} = \frac{1}{\sigma_w^2} \sum_{k=0}^{N-1} \sin^2(2\pi F_0 k) = \frac{N}{2\sigma_w^2}$$

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{N}{\sigma_w^2} & 0 \\ 0 & \frac{N}{2\sigma_w^2} \end{bmatrix} \Rightarrow \text{CRB}(\boldsymbol{\theta}) = \mathbf{I}^{-1}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\sigma_w^2}{N} & 0 \\ 0 & \frac{2\sigma_w^2}{N} \end{bmatrix}$$

$$\text{CRB}(A) = \frac{\sigma_w^2}{N} = \text{MSE}\{\hat{A}_{ML}\}, \quad \text{CRB}(B) = \frac{2\sigma_w^2}{N} = \text{MSE}\{\hat{B}_{ML}\}$$

Hence, the two estimators are efficient.